MILITARY SPECIFICATION SHEET

CONTROL GROUP, ELECTRIC POWER OK-XXX(V)/A

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

The complete requirements for procuring the control group described herein shall consist of this document and the latest issue of MIL-C-81883.

1.2 Classification.

The Control Group described herein shall consist of the equipment and in the quantities listed as follows:

Equipment Name	Quantity	Fig. No.
Processor	_____	2
Crew Control Panel	_____	3
Multiplexer	_____	5
Demultiplexer	_____	6
Multiplexer/Demultiplexer	_____	7
Coupler/Terminator, Data Link	_____	8

FSC 6110

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:37Z
Check the source to verify that this is the current version before use.

GENERAL REQUIREMENTS

3.4.1	Total Weight-Processor	_____
	Crew Control Panel	_____
	Multiplexer	_____
	Demultiplexer	_____
	Multiplexer/Demultiplexer	_____
	Data Link Coupler/Terminator	_____
3.4.2.1	Operational Stability	
	Minimum Stable Life	_____Hrs
3.4.2.2	Operating Life	
	Processor	_____Hrs
	Crew Control Panel	_____Hrs
	Multiplexer	_____Hrs
	Demultiplexer	_____Hrs
	Multiplexer/Demultiplexer	_____Hrs
	Data Link Coupler/Terminator	_____Hrs
	Control Group	_____Hrs
3.4.2.3	Reliability in Mean Time Between Failure	
	MTBF	_____Hrs
3.4.2.4	Probability of Mission Success	
	PMS	_____
	Mission Time	_____Hrs
3.4.7.1	Vibration (Vibration test requirements of MIL-E-5400 specified here).	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.4.7.3	Nuclear Environment	Applicable
	Category I	Applicable
	Category II	Not Applicable
	Electromagnetic Pulse	
	Pulse Duration	2 u sec
	Field Strength	50 kv/m peak
	Neutron Fluence	N Classified
	Prompt Gamma Dose	Classified
	Gamma Dose Rate	Classified
	Total Dose	R Classified

ELECTRICAL REQUIREMENTS

3.4.8 Input Electric Power

3.4.8.1 Input Power (from Each Power Source)

	VA	V	Ph	Hz	pf
Processor					
Crew Control Panel					
Multiplexer					
Demultiplexer					
Multiplexer/Demultiplexer					

3.4.8.2 Power Dissipation (Both Sources Energized)

	With Input and Output Signals In One State	With Input and Output Signals In Zero State
Processor	_____Watts	_____Watts
Crew Control/Display Panel	_____Watts	_____Watts
Multiplexer	_____Watts	_____Watts
Demultiplexer	_____Watts	_____Watts
Multiplexer/Demultiplexer	_____Watts	_____Watts

Source: https://assist.dla.mil -- Downloaded: 2026-02-26 04:37Z
Check the source to verify that this is the current version before use.

3.4.9	Circuit Isolation	
	Data Link (Either wire of shielded twisted pairs) to ground)	_____Ohm,
	Each Multiplexer: Input Signals to ground	_____Ohm,
	Input Signal to all other input signals	_____Ohm,
	Each Demultiplexer: Output signal to ground	_____Ohm,
	Output Signal to all other output signals	_____Ohm,
	Each LRU: Each power input to other power inputs	_____Ohm,
	Each power input to ground	_____Ohm,
3.4.11	Insulation Resistance	_____Ohm,
3.4.13	Equipment Cooling (Cooling provisions must be specified her	
3.5.1.3	Through-Put-Time	_____ms,
3.5.2	Data Link	
	1. Level of redundancy (no. of transmission paths)	_____
	2. Transmission mode (ex. - half duplex)	_____
3.5.2.1.1	Transmission and Stub Cables	
	Characteristic Impedance at 1 Meg Hz	_____Ohms
	Wire Size	_____
	Distributed Capacitance	_____pf/ft
	Conductor Resistance	_____ohms/
	Jacket Material	_____
	Transmission Cable Length	_____ft, m
	Stub Cable Length	_____ft, m
3.5.2.3	Voltage Level	
	LRU Driver Output Voltage	_____volt Peak-to-Peak Ma
	LRU Receiver Threshold Voltage	_____volt Peak-to-Peak Ma

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.2.4 Bit Rate — Rise Time = _____ ± _____ Nano Sec
Fall Time = _____ ± _____ Nano Sec

3.5.2.5 Undetected Bit Error Rate — BER + One error per _____ bits
Noise environment — MIL-STD-461

3.5.2.6.1.4 Mode Code:
(Additional bit designations should be defined here).
Control Word (Bits 11 through 13)

3.5.2.6.1.5 Status Code (Bits 11 through 19)

3.5.3 Data Processor (The data processor shall be defined here if different from that specified in body of spec.)

3.5.3.2.1 Start Up (Note: this value shall be at least _____ ms, Max as long as the start up time for the multiplexer and demultiplexer).

3.5.3.2.2 Operation (If operation differs from that specified in the general specification it shall be defined here).

3.5.3.2.2.1.1 Input/Output Memory
Number of Bits _____ Min

3.5.3.2.2.2 Boolean Processing
Number of Equations _____

Types: Combinational _____
Sequential Without Time Delay _____
Sequential With Time Delay _____
Partial (Hold Internally) _____

Complexity: Number of Variables _____ Average

3.5.3.3 Interface Characteristics
Interface connections shall be as specified in Table I herein.

3.5.3.4 Reliability — MTBF _____ Hr, Min

3.5.3.5 Elapsed Time Recorder
(Requirements for displaying the elapsed time shall be completely defined here).

3.5.4 Crew Control Panel

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.4.2.1 Start Up — Time = _____ ms, Max

3.5.4.2.2 Operation

Special requirements for compiling and transmitting data to the control panel shall be defined here.

Control panel display shall be as defined by Figure 4 and the associated notes.

3.5.4.3 Interface Characteristics

Interface connections shall be as specified in Table II herein.

3.5.4.4 Reliability — MTBF _____ Hr, Min

3.5.4.5 Elapsed Time Recorder

(Requirements for Displaying the elapsed time shall be completely defined here).

3.5.5 Multiplexer

3.5.5.2.1 Start Up — Time = _____ ms, Max

3.5.5.2.2 Operation (If operation differs from that specified in the general specification it shall be defined here).

3.5.5.3 Interface Characteristics

Interface connections shall be as specified in Table III herein.

3.5.5.3.1 Data Input Signal Characteristics

ON state - (1) _____ V _____ ma

OFF state - (0) _____ V _____ ma

Rise Time - _____ ± _____ nano sec

Fall Time - _____ ± _____ nano sec

3.5.5.4 Reliability — MTBF _____ Hr, Min

3.5.5.5 Elapsed Time Recorder

(Requirements for displaying the elapsed time shall be completely defined here).

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.6 Demultiplexer

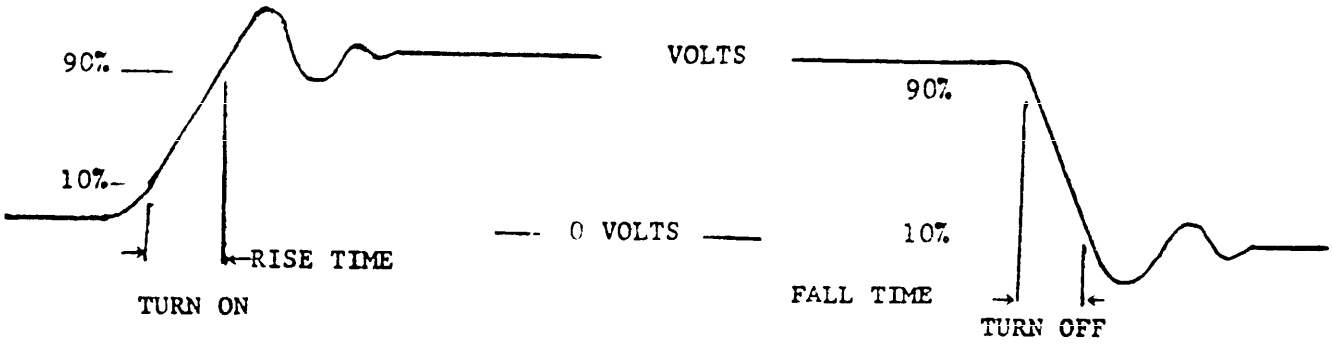
3.5.6.2.1 Start Up Time = _____ ms, Max

3.5.6.2.2 Operation (If operation differs from that specified in the general specification it shall be defined here).

3.5.6.3 Interface Characteristics
Interface connections shall be as specified in Table IV herein.

3.5.6.3.1 Driver Signal Characteristics

Logic "1" State	_____ V d.c., _____ ma
Logic "0" State	_____ V d.c., _____ ma
Rise Time	10 Microsec Max
Fall Time	10 Microsec Max



3.5.6.4 Reliability MTBF _____Hr, Min

3.5.6.5 Elapsed Time Recorder
(Requirements for displaying the elapsed time shall be completely defined here).

3.5.7 Multiplexer/Demultiplexer

3.5.7.2.1 Start Up Time = _____ms, Min

3.5.7.3 Interface Characteristics
Interface connections shall be as specified in Table V herein

3.5.7.4 Reliability MTBF _____Hr, Min

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.7.5	Elapsed Time Recorder (Requirements for displaying the elapsed time shall be completely defined here).	
3.5.8	Data Link Coupler/Terminator	
3.5.8.1	Function	
	Coupling Network	Fig. 9
	Termination Network	Fig. 9
3.5.8.2	Operation	
	Coupling Network Impedance	
	(a) To the Transmission Cable (at _____Hz)	_____Ohms
	(b) To the Stub Cable	_____Ohms
3.5.8.3	Interface Characteristics Interface connections shall be as specified in Table VI herein.	
3.5.8.4	Reliability	MTBF _____Hr, Min
3.5.9	Built-In-Test	
3.5.9.2.3	BIT Test Times	
	(a) Fault Detection, Update Intervals	_____ ms, Max
3.5.10	Software	Language _____
3.5.10.1	Application Software	Language _____
3.5.10.2	Automatic Test Equipment Software	Language _____
3.5.10.3	Support Software	Language _____
3.5.10.4	Software Documentation	Listing of Programs Submitted

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:37Z
Check the source to verify that this is the current version before use.

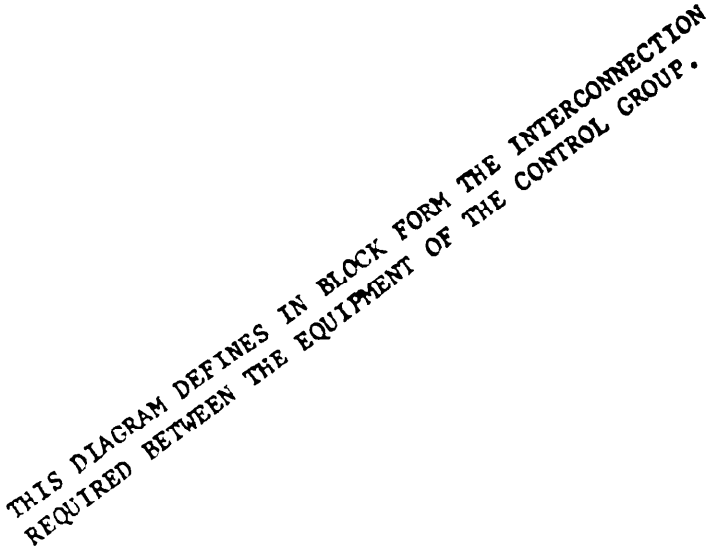


FIGURE 1. A System Block Diagram for the Control Group, Electric Power, Aircraft

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:37Z
Check the source to verify that this is the current version before use.

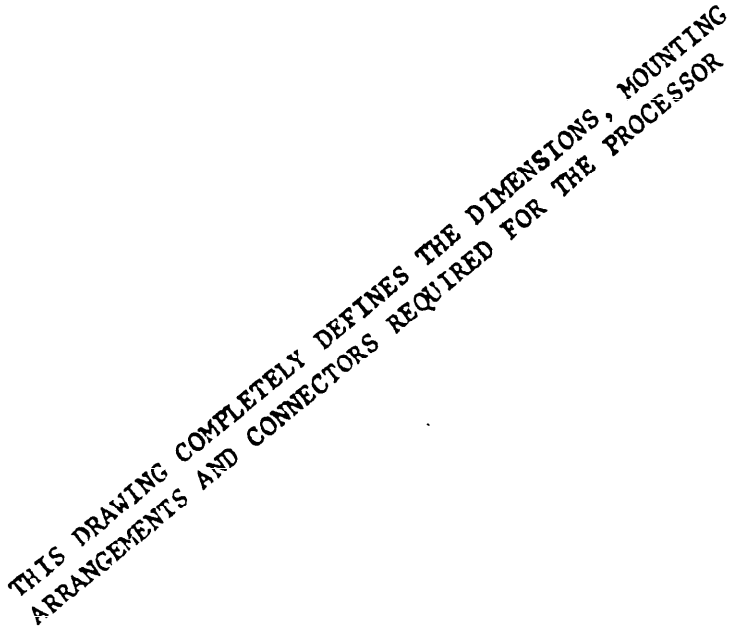


FIGURE 2. Processor Outline and Mounting Drawing

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

TABLE I

PROCESSOR CONNECTOR PIN ASSIGNMENT

i.e. DATA LINK CONNECTOR (J____) (CONNECTOR DESIGNATOR)

CONNECTOR PART NUMBER:

PIN NUMBER	FUNCTION ASSIGNMENT

A TABLE SHALL BE PROVIDED FOR EACH CONNECTOR TO SHOW THE PIN NUMBER AND FUNCTION ASSIGNMENT (WHERE SPACE PERMITS, SEVERAL TABLES MAY BE PLACED ON THE SAME PAGE)

NOTES:

1. Redundant lines shall not be included in the same connector.
2. Test point connectors shall conform to para. 3.4.5.2.
3. Connectors used for Memory Loading are for Ground Equipment connection only and shall be provided with a protective captive cover for use when installed in the aircraft.
4. Explanatory notes and special instructions (if required) shall be shown for each Table, i.e.:
    1. None of the pinslisted should ever be connected to external voltage.
    2. Pins X, X, X, should never be grounded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

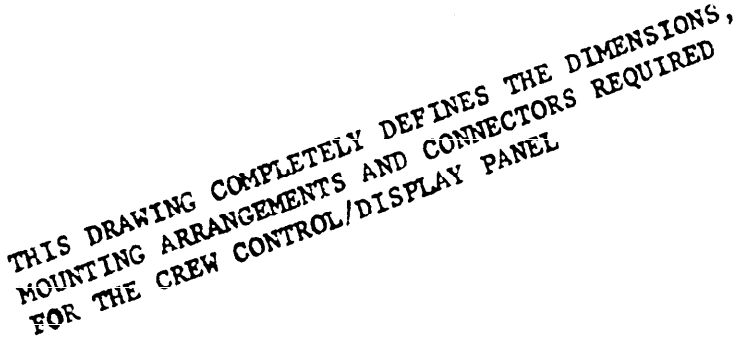


FIGURE 3. Crew Control Panel Outline and Mounting Drawing

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:37Z
Check the source to verify that this is the current version before use.

TABLE II

CREW CONTROL PANEL

CONNECTOR PIN ASSIGNMENT

i.e. DATA LINK CONNECTOR (J____)(CONNECTOR DESIGNATOR)

CONNECTOR PART NUMBER: (XXXXXXXX)

PIN NUMBER	FUNCTION ASSIGNMENT

A TABLE SHALL BE PROVIDED FOR EACH CONNECTOR TO SHOW THE PIN NUMBER AND FUNCTION ASSIGNMENT (WHERE SPACE PERMITS, SEVERAL TABLES MAY BE PLACED ON THE SAME SHEET)

NOTES:

1. Redundant lines shall not be included in the same connector.
2. Test Point Connectors shall conform to paragraph 3.4.5.2.
3. Explanatory notes and special instructions (if required) shall be shown for each table i.e.:

    1. None of the pins listed should ever be connected to external voltage.
    2. Pins X,X,X should never be grounded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

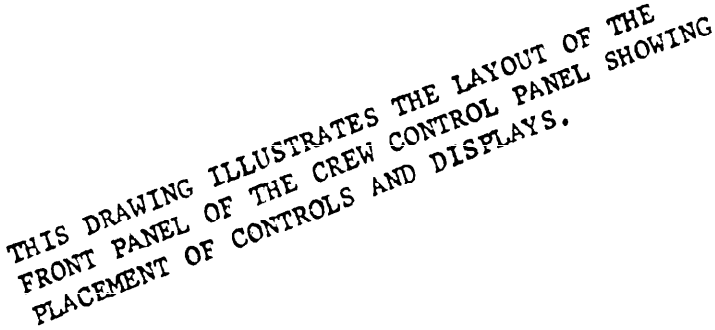


FIGURE 4. Crew Control Panel - Front Panel Layout

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:37Z
Check the source to verify that this is the current version before use.

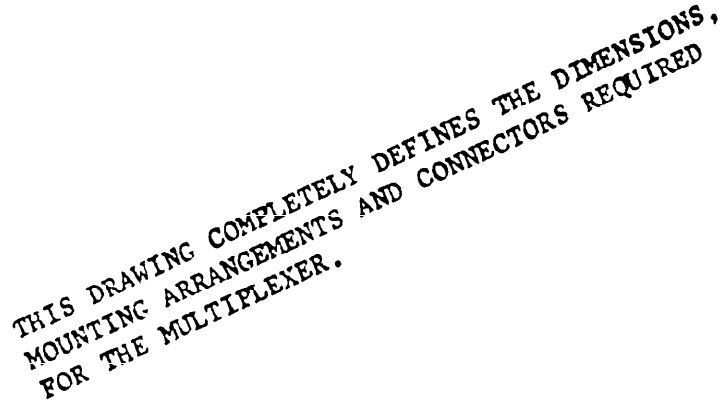


FIGURE 5. Multiplexer Outline and Mounting Drawing

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:37Z.
Check the source to verify that this is the current version before use.

TABLE III

MULTIPLEXER CONNECTOR PIN ASSIGNMENT

i.e., DATA LINK CONNECTOR (J____) (CONNECTOR DESIGNATOR)

CONNECTOR PART NUMBER: (XXXXXXXXXXXXXXXX)

PIN NUMBER	FUNCTION ASSIGNMENT

A TABLE SHALL BE PROVIDED FOR EACH CONNECTOR TO SHOW THE PIN NUMBER AND FUNCTION ASSIGNMENT (WHERE SPACE PERMITS, SEVERAL TABLES MAY BE PLACED ON THE SAME PAGE)

NOTES:

1. Redundant lines shall not be included in the same connector.
2. Test point connectors shall conform to para. 3.4.5.2.
3. Explanatory notes and special instructions (if required) shall be shown for each table, i.e.:
    1. None of the pins listed should ever be connected to external voltage.
    2. Pins X, X, X, should never be grounded.

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:37Z
Check the source to verify that this is the current version before use.

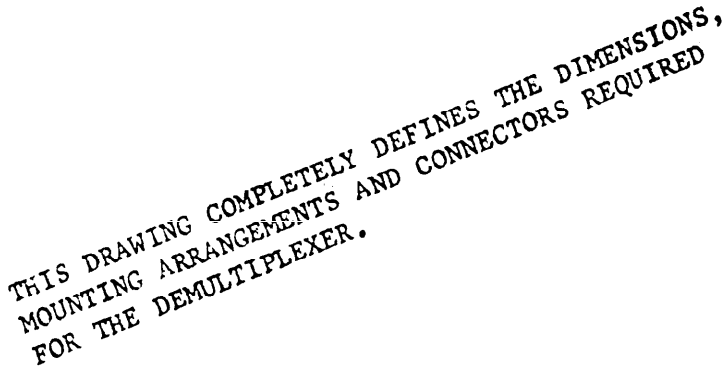


FIGURE 6. Demultiplexer Outline and Mounting Drawing

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:37Z
Check the source to verify that this is the current version before use.

TABLE IV

DEMULTIPLEXER CONNECTOR PIN ASSIGNMENT

i.e., DATA LINK CONNECTOR (J___) (CONNECTOR DESIGNATOR)

CONNECTOR PART NUMBER: (XXXXXXXXXXXX)

PIN NUMBER	FUNCTION ASSIGNMENT

A TABLE SHALL BE PROVIDED FOR EACH CONNECTOR TO SHOW THE PIN NUMBER AND FUNCTION ASSIGNMENT (WHERE SPACE PERMITS, SEVERAL TABLES MAY BE PLACED ON THE SAME PAGE)

NOTES:

1. Redundant lines shall not be included in the same connector.
2. Test point connectors shall conform to para. 3.4.5.2.
3. Explanatory notes and special instructions (if required) shall be shown for each table, i.e.:
    1. None of the pins listed should ever be connected to external voltage.
    2. Pins X, X, X, should never be grounded.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:37Z
Check the source to verify that this is the current version before use.

THIS DRAWING COMPLETELY DEFINES THE DIMENSIONS, MOUNTING ARRANGEMENTS AND CONNECTORS REQUIRED FOR THE MULTIPLEXER/DEMULTIPLEXER.

FIGURE 7. MUX/DEMUX OUTLINE AND MOUNTING DRAWING

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

TABLE V

MULTIPLEXER/DEMULTIPLEXER CONNECTOR PIN ASSIGNMENT

i.e., DATA LINK CONNECTOR (J___) (CONNECTOR DESIGNATOR)

CONNECTOR PART NUMBER: (XXXXXXXXX)

PIN NUMBER	FUNCTION ASSIGNMENT

A TABLE SHALL BE PROVIDED FOR EACH CONNECTOR TO SHOW THE PIN NUMBER AND FUNCTION ASSIGNMENT (WHERE SPACE PERMITS, SEVERAL TABLES MAY BE PLACED ON THE SAME PAGE).

NOTES:

1. Redundant lines shall not be included in the same connector
2. Test point connectors shall conform to para 3.4.5.2
3. Explanatory notes and special instructions (if required) shall be shown for each table i.e.:
    1. None of the pins listed should ever be connected to external voltage.
    2. Pins X, X, X, should never be grounded.

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:37Z
Check the source to verify that this is the current version before use.

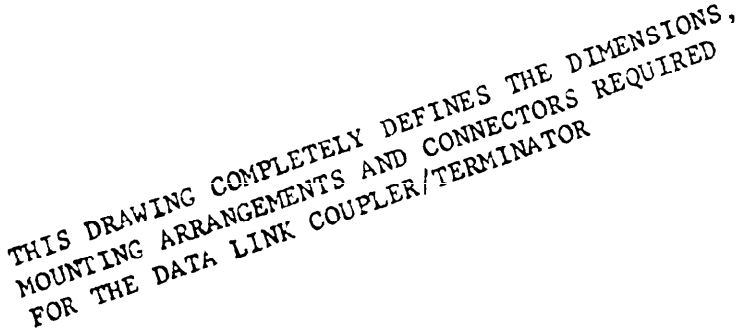


FIGURE 8. Data Link Coupler/Terminator Outline and Mounting Drawing

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:37Z
Check the source to verify that this is the current version before use.

TABLE VI

DATA LINK COUPLER/TERMINATOR

CONNECTOR PIN ASSIGNMENT

i.e., DATA LINK CONNECTOR (J___) (CONNECTOR DESIGNATOR)

CONNECTOR PART NUMBER (XXXXXXXX)

PIN NUMBER	FUNCTION ASSIGNMENT

A TABLE SHALL BE PROVIDED FOR EACH CONNECTOR TO SHOW THE PIN NUMBER AND FUNCTION ASSIGNMENT (WHERE SPACE PERMITS, SEVERAL TABLES MAY BE PLACED ON THE SAME PAGE).

NOTES:

1. Redundant lines shall not be included in the same connector
2. Test point connectors shall conform to para. 3.4.5.2
3. Explanatory notes and special instructions (if required) shall be shown for each table i.e.:

    1. None of the pins listed should ever be connected to external voltage.
    2. Pins X, X, X, should never be grounded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

THIS DRAWING SHOWS THE INTERNAL COMPONENTS AND CONNECTIONS (SCHEMATIC REPRESENTATION) OF THE DATA LINK COUPLER/TERMINATOR

FIGURE 9. Data Link Coupler/Terminator Internal Connections

Preparing Activity
Navy - AS

Project No. 6110-N203

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:37Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers / vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-C-81383/1(AS) CONTROL GROUP, ELECTRIC POWER OK-XXX(V)/A

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-04-26T04:37Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

Naval Air Engineering Center
Lakehurst, NJ 08733

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID

U.S. MAIL

Engineering Specification and Standards Department (Code 93)
Naval Air Engineering Center
Lakehurst, NJ 08733

FOLD

Source: https://assist.dla.mil -- Downloaded: 20260126104 372
Check the source to verify that this is the current version before use.